

FIT5047 Assignment 1 Installation Guide

Version 1.3

2026

This document provides supporting documentation to help you come to grips with the Pac-Man game environment which we will use for Assignment 1.

Required Skills

- Programming with Python
- Using git to manage codes
- Using command line interface

No worries if some of these skills are new to you - that's totally okay. These links right here are your pathway to learning. They're here to guide you at your own pace, leading you step-by-step towards mastery. Dive in and out of these resources as you feel appropriate.

- Python introduction:
https://www.w3schools.com/python/python_intro.asp
- Git introduction:
<https://github.com/git-guides>
- Linux/Mac CLI (Command Line Interface) Guide:
<https://ubuntu.com/tutorials/command-line-for-beginners#1-overview>
- Windows CMD cheat sheet:
<http://www.cs.columbia.edu/~sedwards/classes/2015/1102-fall/Command%20Prompt%20Cheatsheet.pdf>
- The MIT Missing Semester in Computer Science:
<https://missing.csail.mit.edu/>

Installation and Setup

Step 1: Create a GitHub account

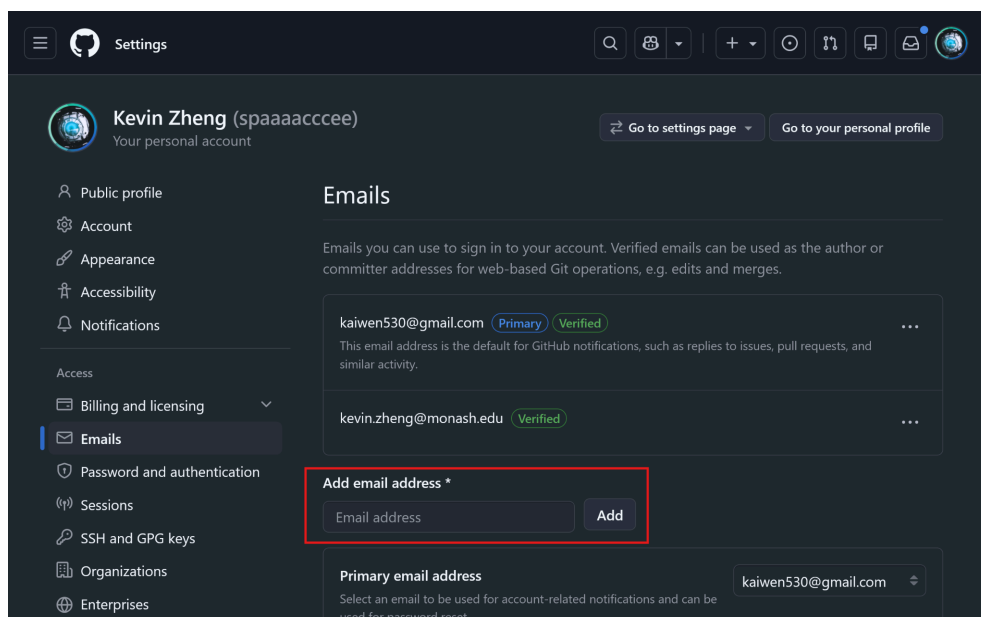
For this assignment you will upload your code to a GitHub repository from where it will be automatically evaluated.

Go to [GitHub's sign up page](#) and follow the steps to create a GitHub account. When prompted for your email, enter your **Monash email address**. **The competition system will not allow you to submit from a non-Monash email address.**

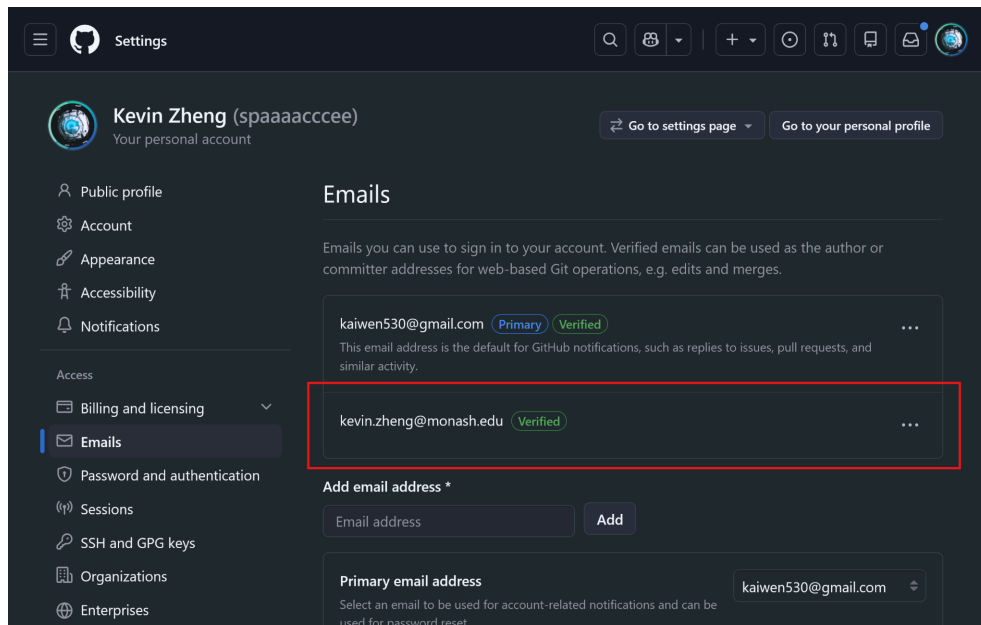
Alternative: Use an existing GitHub account

We **highly recommend** that you create a dedicated GitHub account using your student email for your assignments. If you already own and wish to reuse a personal GitHub account, you may use that one by associating your student email with it:

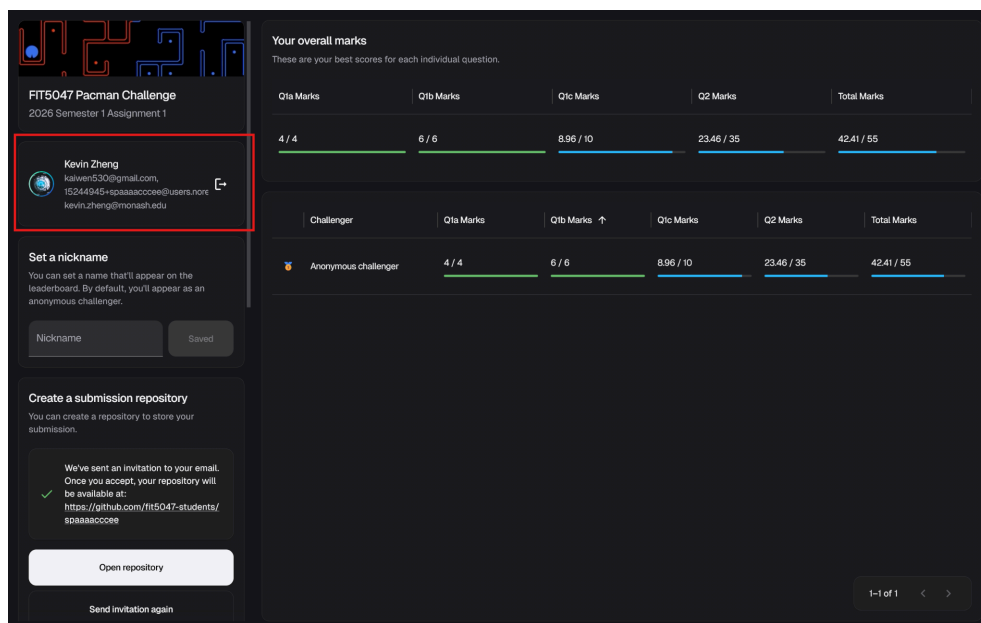
Go to your [GitHub account email settings](#). In Add email address, type in your Monash email address and press Add.



Follow the steps GitHub prompts you to verify your email address. Check that your email has appeared in the following list.



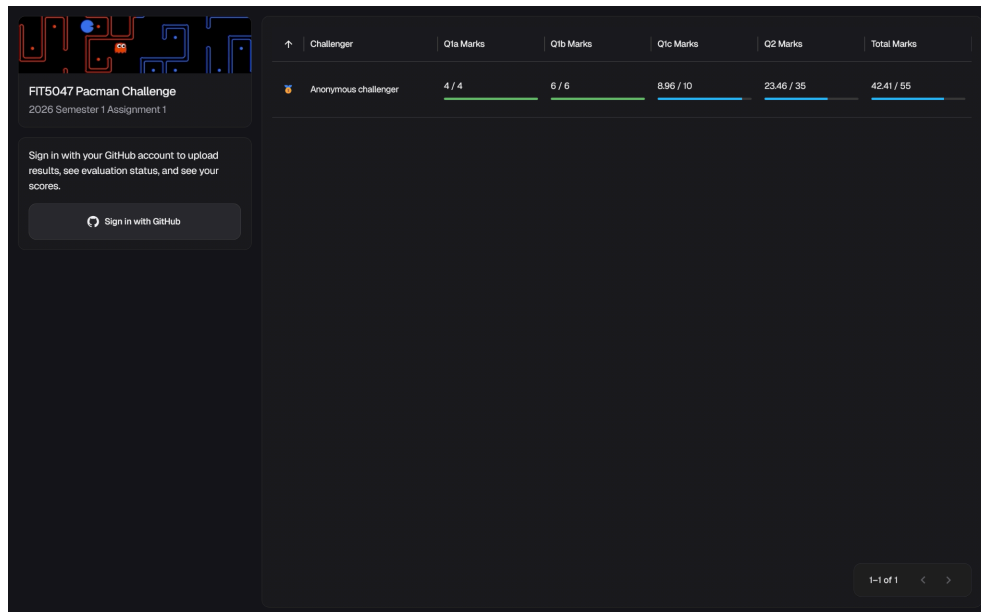
Go to the [competition site](#). After logging in (see below), check that your email has appeared in this list. You may have to wait a few minutes before it updates.



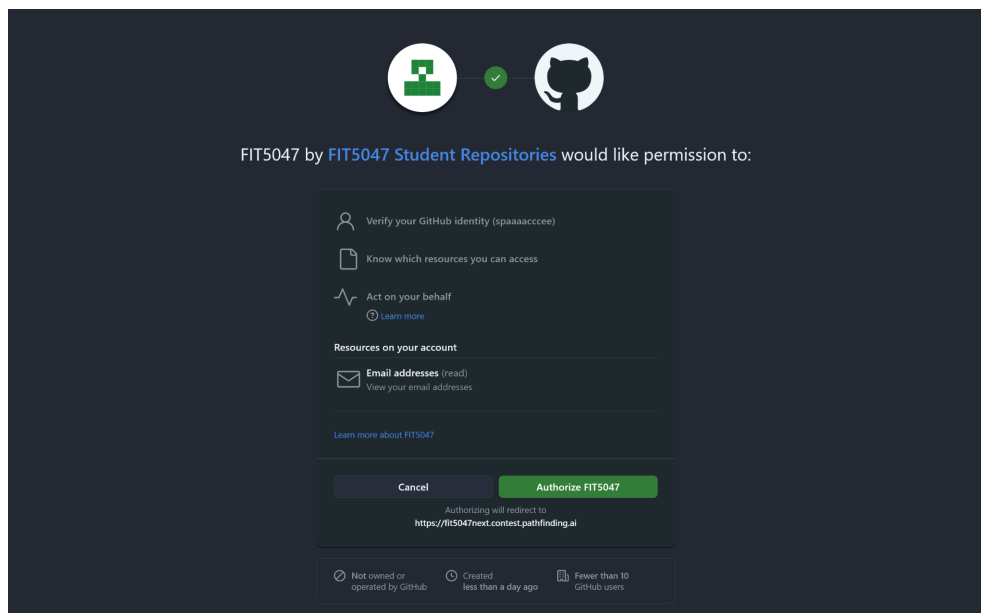
Step 2: Sign in to the auto-grader server

Our auto-grading server evaluates your submissions and lets you compare your progress in the assignment against your classmates, using a live leaderboard. You can find the auto-grader website here: [FIT5047 Pacman Challenge](#).

Click **Sign in with GitHub** to sign in with your GitHub account.



You'll be taken to an authorization screen. Click **Authorize FIT5047**.



You'll be taken back to the auto-grader. Verify that you can see your name, avatar and email addresses.

The screenshot displays the FIT5047 Pacman Challenge interface. On the left sidebar, the user is identified as Kevin Zheng with email kaiven530@gmail.com and 15244945+spaaacocoe@users.nore. The interface includes sections for setting a nickname, creating a submission repository (with a GitHub link), and a leaderboard titled 'Your overall marks'. The leaderboard shows scores for Q1a, Q1b, Q1c, Q2, and Total Marks for the user and an anonymous challenger.

Your overall marks					
These are your best scores for each individual question.					
Q1a Marks	Q1b Marks	Q1c Marks	Q2 Marks	Total Marks	
4 / 4	6 / 6	8.96 / 10	23.46 / 35	42.41 / 55	

Challenger	Q1a Marks	Q1b Marks	Q1c Marks	Q2 Marks	Total Marks
Anonymous challenger	4 / 4	6 / 6	8.96 / 10	23.46 / 35	42.41 / 55

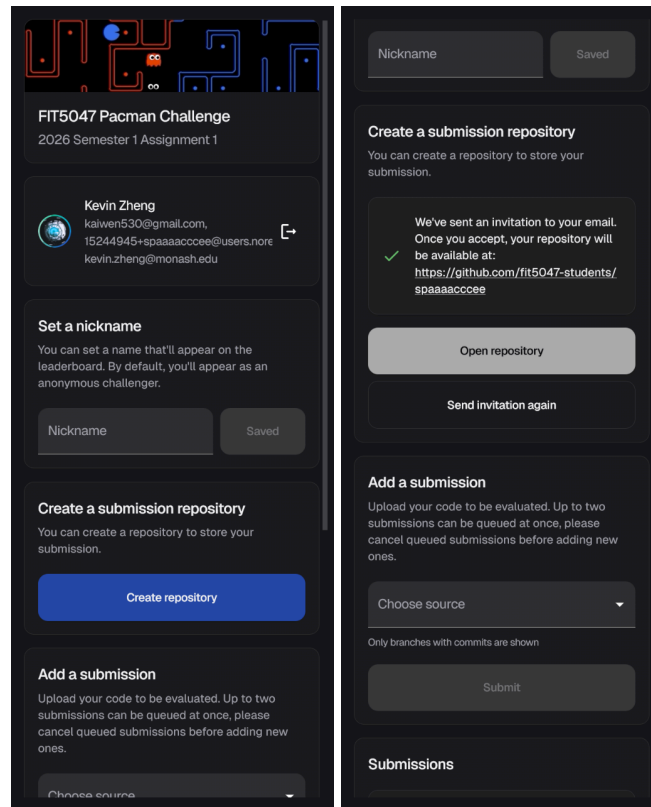
You are now signed in.

Switching accounts

If you sign out of the auto-grader, GitHub will remember your account so next time you sign in it'll be promptless. If you wish to switch accounts, you must first sign out of both the auto-grader and GitHub. Then, you may sign in again as usual.

Step 3: Prepare your submission repository

We'll create a repository that'll hold the code that you'll be graded on. Click **Create repository**. After a little while, you'll see that your repository has been initialised.



There are multiple ways to accept your invitation:

Go directly to the repository

If you navigate to your repository by clicking on "Open repository", you should be presented with a prompt to accept your invitation.

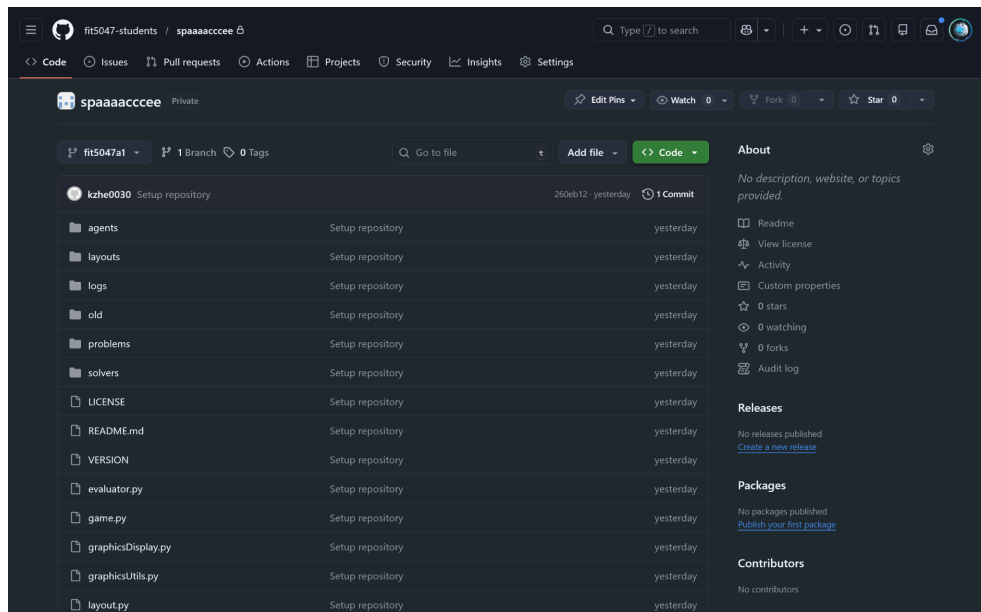
Find it in GitHub notifications

You may also find the invitation on your [GitHub notifications page](#).

Find it in your inbox

An email will be sent to the primary email address associated with your GitHub account. This would be the first email address shown next to your avatar on the auto-grader.

Accept the invitation to access your submission repository. It'll be prepopulated with starter code.



Step 4: Create a development environment

For this assignment we will implement our agents using the Python programming language. We also recommended using miniconda, a lightweight tool that helps us install Python, and manage different Python environments.

Install Conda

Follow [this guide](#) to install miniconda. Please refer to the appropriate instructions for your operating system. Alternatively, you can use [anaconda](#) which provides more features than miniconda, such as a GUI interface.

Once miniconda is installed, type the following commands into your terminal:

```
$ conda create -n fit5047_pacman python=3.9
$ conda activate fit5047_pacman
$ conda install -c anaconda tk
```

Install Git

If you don't already have the git version management software installed, you will need to do that now. The easiest approach is to install git via miniconda:

```
$ conda install anaconda::git
```

Confirm that git successfully installed by checking its version.

```
$ git --version
```

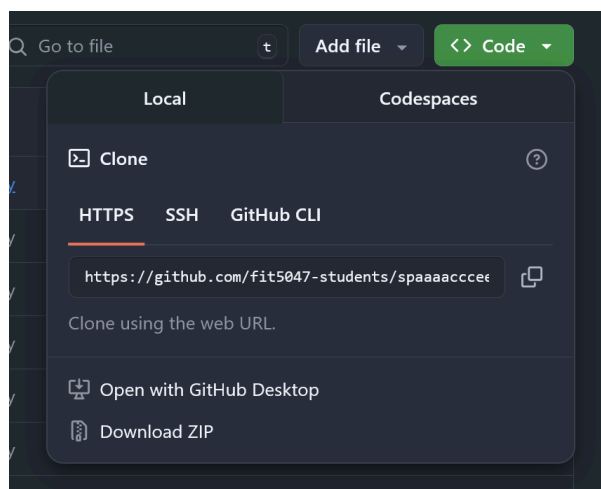
Clone the repo to your computer

Use the terminal and navigate to a directory in which you would like to clone the Pac-Man repository. For example, from your home directory:

```
$ mkdir fit5047  
$ cd fit5047
```

For a refresher on navigating with a terminal interface, you may like to consult the [MIT Missing Semester in Computer Science](#), which is a terrific resource for learning not just about the terminal interface, but also the git and a variety of command line tools.

Next, from the "source" page of your GitHub repository, click the **Code** button, copy the "Clone" command shown and paste it into the terminal.



There are two methods to clone your code, HTTPS and SSH.

CLONE USING HTTPS

When cloning with HTTPS, the command should look something like this:

```
$ git clone https://github.com/fit5047-students/<YOUR_USERNAME>.git
```

If you use HTTPS to clone you will be prompted to enter your GitHub username and password. You can also [create a GitHub access token](#) to use to authenticate in place of your password.

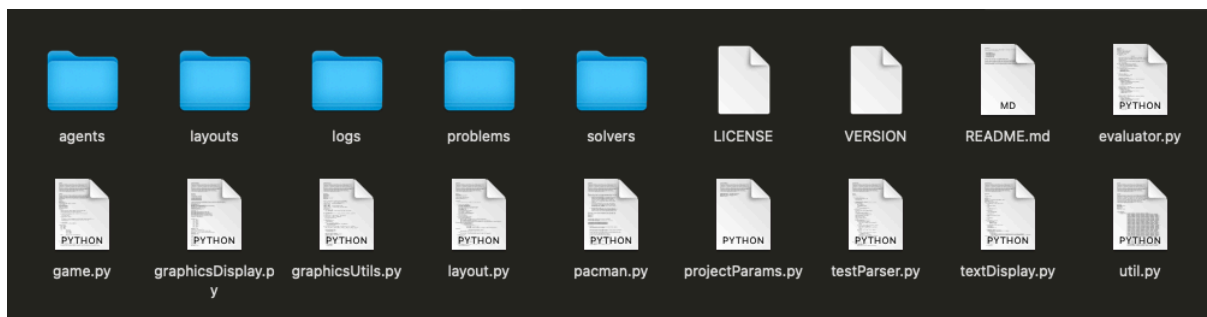
CLONE USING SSH

If HTTPS doesn't work for you, you can try to use SSH to clone your repo instead. You need to [create an SSH key](#) and [add it to your GitHub account](#). Then you can clone your repo using:

```
$ git clone git@github.com:fit5047-students/<YOUR_USERNAME>.git
```

If that still doesn't work, then please post on Ed or attend one of the consultations.

Once the command finishes you will have a new subdirectory called <YOUR_USERNAME>, in your current directory. The contents of the <YOUR_USERNAME> directory should look like this:



The agents, layouts, problems and solvers directories are all important for you. We will go through each of these directories in more detail in the "Getting Started Guide" (soon to be released). **You shouldn't modify any of the python files in this directory.**

Additional support software (optional)

Many programmers like to write code in an Integrated Development Environment (IDE). If you want to try an IDE but aren't sure which, we suggest [VS Code](#).

Another popular (optional) tool is [Jupyter notebook](#), which you can use for your own testing and development. Note that all files you submit must be the specified .py files (i.e., please do not submit your Jupyter notebook!).

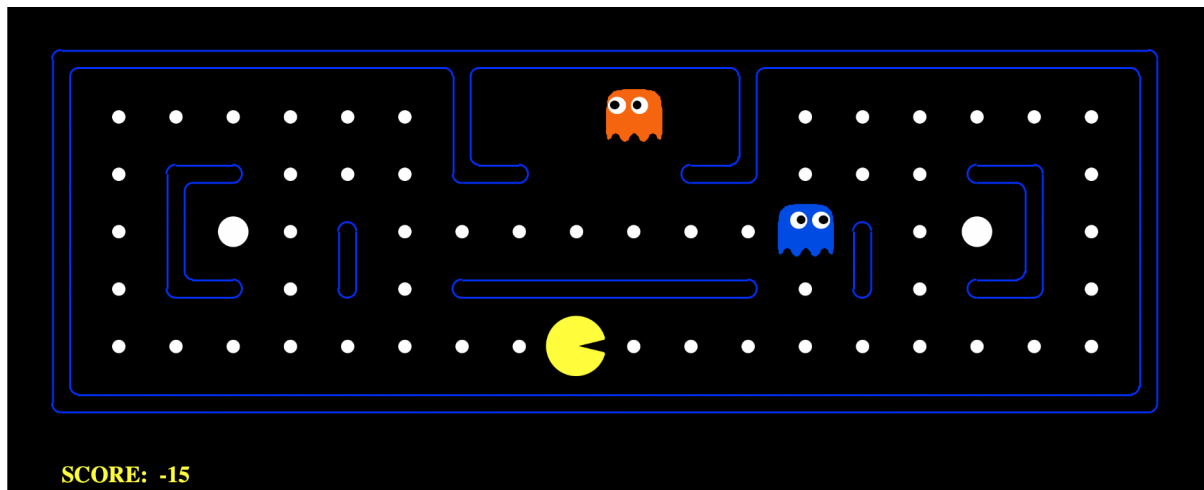
Test your setup

Once you have completed the above Instructions we make sure the environment is running correctly. We want to bring up the visualiser and run the game with two commands to verify the installation was successful. Make sure you are in the directory where the pacman.py file is located then enter the commands below in the terminal. Each game will end when Pac-Man has eaten all the food on the map, or when Pac-Man is eaten by a ghost.

Demo with a keyboard controller:

```
$ python pacman.py -l smallClassic.lay
```

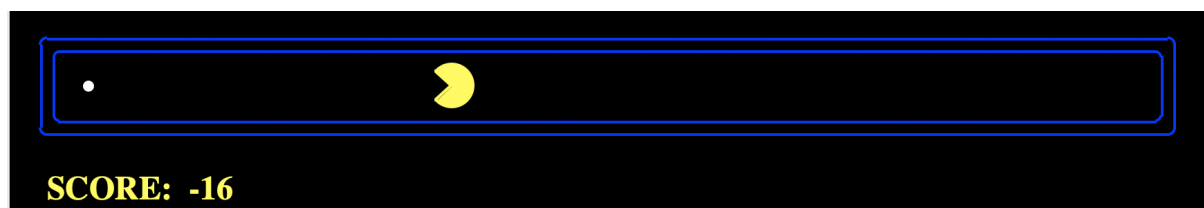
This command will begin a new game and allow you to control Pac-Man with the WASD keys on your keyboard.



Demo with a very simple agent:

```
$ python pacman.py --layout testMaze.lay --pacman GoWestAgent
```

This command will begin a new game where the automated controller only ever tells Pac-Man to "Go West".



You can use the -o flag to specify the file name for saving a log file in the logs directory. You can inspect these log files for debugging purposes but you should not modify them.

Making a submission

You should be “committing” and “pushing” your code to the submission repository. A commit saves your work to a local Git history. In a terminal, you may commit by:

```
$ git add *  
$ git commit -m "A message to help you keep track of changes"
```

A push will upload all your commits to the remote repository, in this case the GitHub repository owned by the auto-grader:

```
$ git push
```

After you push, verify that your push was successful by checking it was reflected in the GitHub source page. The commit should also appear in the auto-grader. Select the commit and click **Submit** to finalise your submission.

Send invitation again

Add a submission
Upload your code to be evaluated. Up to two submissions can be queued at once, please cancel queued submissions before adding new ones.

Choose source
spaaaaaccee fit5047a1 268eb12

Only branches with commits are shown

Repository
spaaaaaccee

Branch
fit5047a1

Commit
kzhe0030, 1 day ago, Setup repository

Commit hash
268eb12

Submit

Submissions

fit5047-students/spaaaaaccee
example1
Done, 1 of 2 checks failed

Your overall marks
These are your best scores for each individual question.

Q1a Marks	Q1b Marks	Q1c Marks	Q2 Marks	Total Marks
4 / 4	6 / 6	8.96 / 10	23.46 / 35	42.41 / 55

Challenger	Q1a Marks	Q1b Marks	Q1c Marks	Q2 Marks	Total Marks
Anonymous challenger	4 / 4	6 / 6	8.96 / 10	23.46 / 35	42.41 / 55

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To keep it fair, the auto-grader will only allow you to make one submission at a time. Press the bin icon on any pending or running submission to cancel them.

More Information

- Read the Assignment Specification and Marking Rubric (soon to be released on Moodle)
- Read the Marking Rubric again, and make sure you really understand what is expected.
- Check out the "Getting Started Guide" (soon to be released on Moodle) for descriptions of the Pac-Man game environment and tips for how to get started with the assignment.

Revision History

Version 1.1

- Updated git clone command from git clone `https://github.com/fit5047-students/<YOUR_USERNAME>.git` to git clone https://github.com/FIT5047-sem2-2025-students/<YOUR_USERNAME>.git

Version 1.2

- Added more detail to the SSH git clone instructions

Version 1.3

- Added instructions for new server system

Version 1.3.1

- Removed an errant "z" from miniconda installation instructions (page 7)